Assignment -03

Course Details

Course Title: Computer Networks

Couse Code: CSE 405

Section: 05

Submitted To

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Network Topology Summary

The network contains:

- **Three LANs**
  - LAN 1 → 190.110.0.0
  - LAN 2 → 122.0.0.0
  - LAN 3 → 222.0.0.0
- Three routers connected using serial links
- Three switches, one for each LAN
- Six PCs two PCs in each LAN
- Two serial networks used between routers:
  - 120.0.0.0 (Router0 to Router1)
  - 220.90.90.0 (Router1 to Router2)

IP Addressing Table

Device / Interface	IP Address	Subnet Mask	Role
PC0 Fa0	190.110.0.1	255.255.0.0	Host
PC1 Fa0	190.110.0.2	255.255.0.0	Host
Router0 Fa0/0	190.110.0.254	255.255.0.0	LAN DG
Router0 Se2/0	120.0.0.1	255.0.0.0	to Router1
Router1 Se2/0	120.0.0.2	255.0.0.0	to Router0
PC2 Fa0	122.0.0.1	255.0.0.0	Host
PC3 Fa0	122.0.0.2	255.0.0.0	Host
Router1 Fa0/0	122.0.0.254	255.0.0.0	LAN DG
Router1 Se3/0	220.90.90.1	255.255.255.0	to Router2
Router2 Se2/0	220.90.90.2	255.255.255.0	to Router1
PC4 Fa0	222.0.0.1	255.255.255.0	Host
PC5 Fa0	222.0.0.2	255.255.255.0	Host
Router2 Fa0/0	222.0.0.254	255.255.255.0	LAN DG

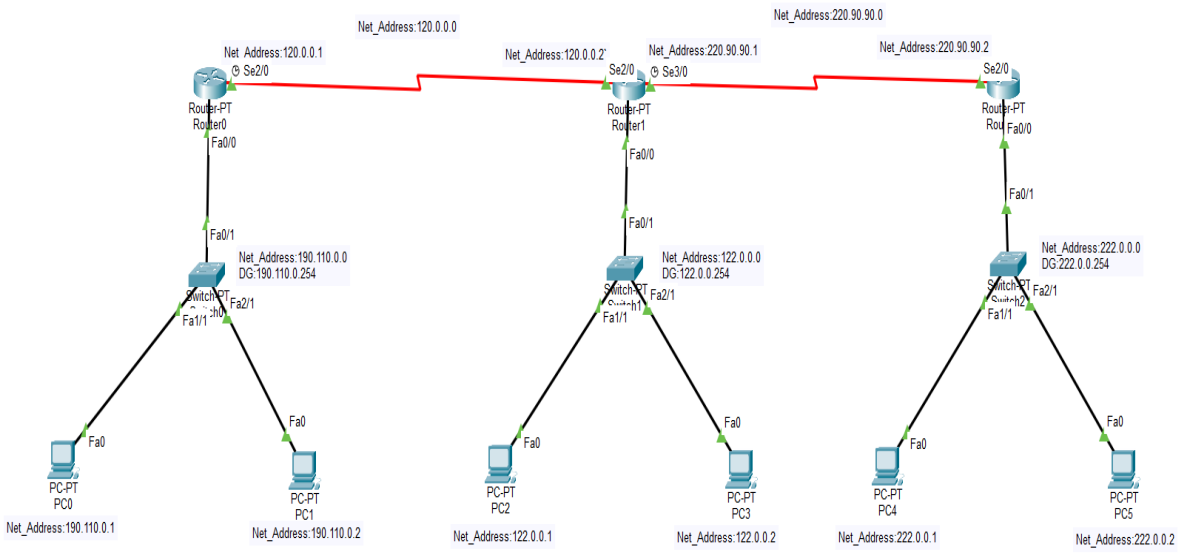


Figure: Network diagram with OSPF, RIP and EIGRP Router Protocol

Router Configurations

For Router 0:

fa0/0

```
Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip address 190.110.0.254 255.255.0.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

se2/0

```
Router(config)#int se2/0
Router(config-if)#ip address 120.0.0.1 255.0.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

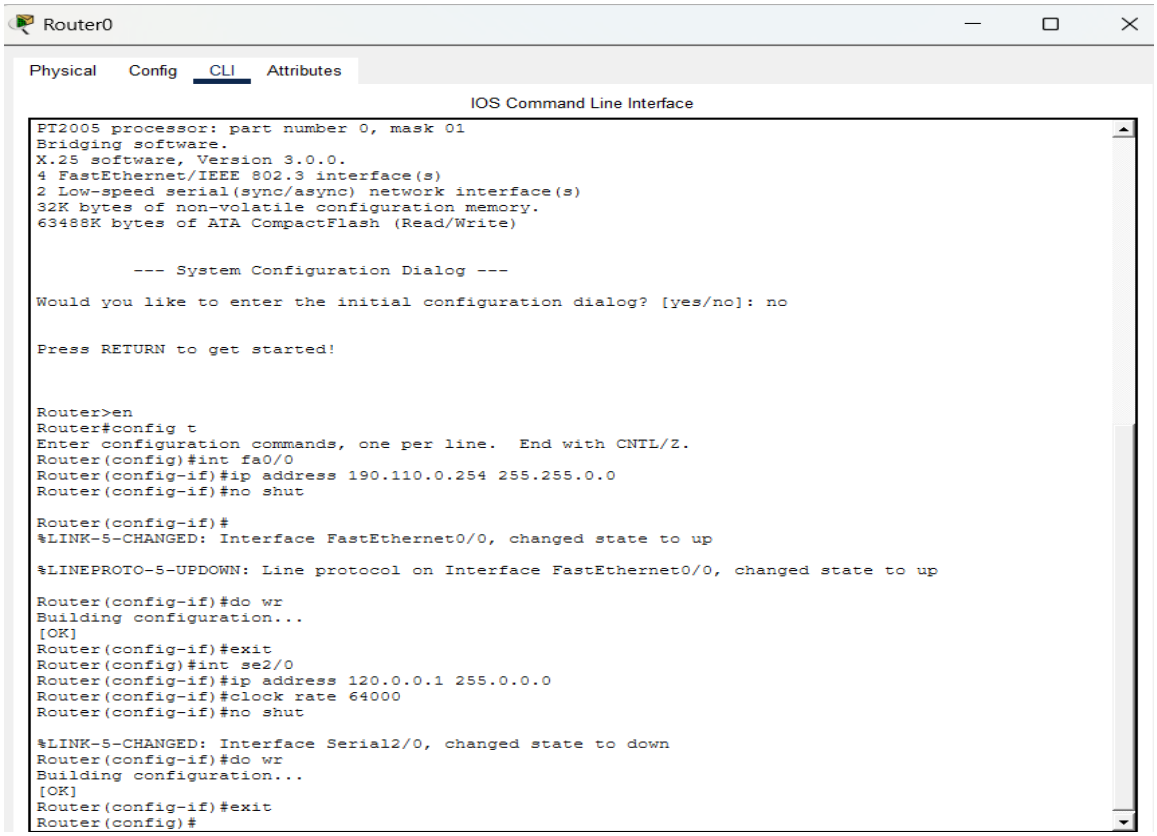


Figure: Router0 configuration

Router0 (RIP Configuration)

Router(config)#router rip
Router(config-router)#network 120.0.0.0
Router(config-router)#network 190.110.0.0
Router(config-router)#exit

```
Router(config)#router rip
Router(config-router)#network 120.0.0.0
Router(config-router)#network 190.110.0.0
Router(config-router)#exit
```

Router0 (OSPF Configuration)

Router(config-router)#network 120.0.0.0 0.255.255.255 area 1
Router(config-router)#network 190.110.0.0 0.0.255.255 area 1
Router(config-router)#exit

```
Router(config)#router ospf 1
Router(config-router)#network 120.0.0.0 0.255.255.255 area 1
Router(config-router)#network 190.110.0.0 0.0.255.255 area 1
Router(config-router)#exit
```

Router0 (EIGRP Configuration)

Router(config-router)#network 120.0.0.0 0.255.255.255

Router(config-router)#network 190.110.0.0 0.0.255.255
Router(config-router)#exit

```
Router(config)#router eigrp 1
Router(config-router)#network 120.0.0.0 0.255.255.255
Router(config-router)#network 190.110.0.0 0.0.255.255
Router(config-router)#exit
```

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router con0 is now available



Press RETURN to get started.



Router>en
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#network 120.0.0.0
Router(config-router)#network 190.110.0.0
Router(config-router)#exit
Router(config)#router ospf 1
Router(config-router)#network 120.0.0.0 0.255.255.255 area 1
Router(config-router)#network 190.110.0.0 0.0.255.255 area 1
Router(config-router)#exit
Router(config)#router eigrp 1
Router(config-router)#network 120.0.0.0 0.255.255.255
Router(config-router)#network 190.110.0.0 0.0.255.255
Router(config-router)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```

Figure: Router0 configuration with dynamic routing protocols OSPF, RIP and EIGRP

For Router 1:

se2/0

Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int se2/0

```
Router(config-if)#ip address 120.0.0.2 255.0.0.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

fa0/0

```
Router(config)#int fa0/0
Router(config-if)#ip address 122.0.0.254  255.0.0.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

se3/0

```
Router(config)#int se3/0
Router(config-if)#ip address 220.90.90.1 255.255.255.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit
```

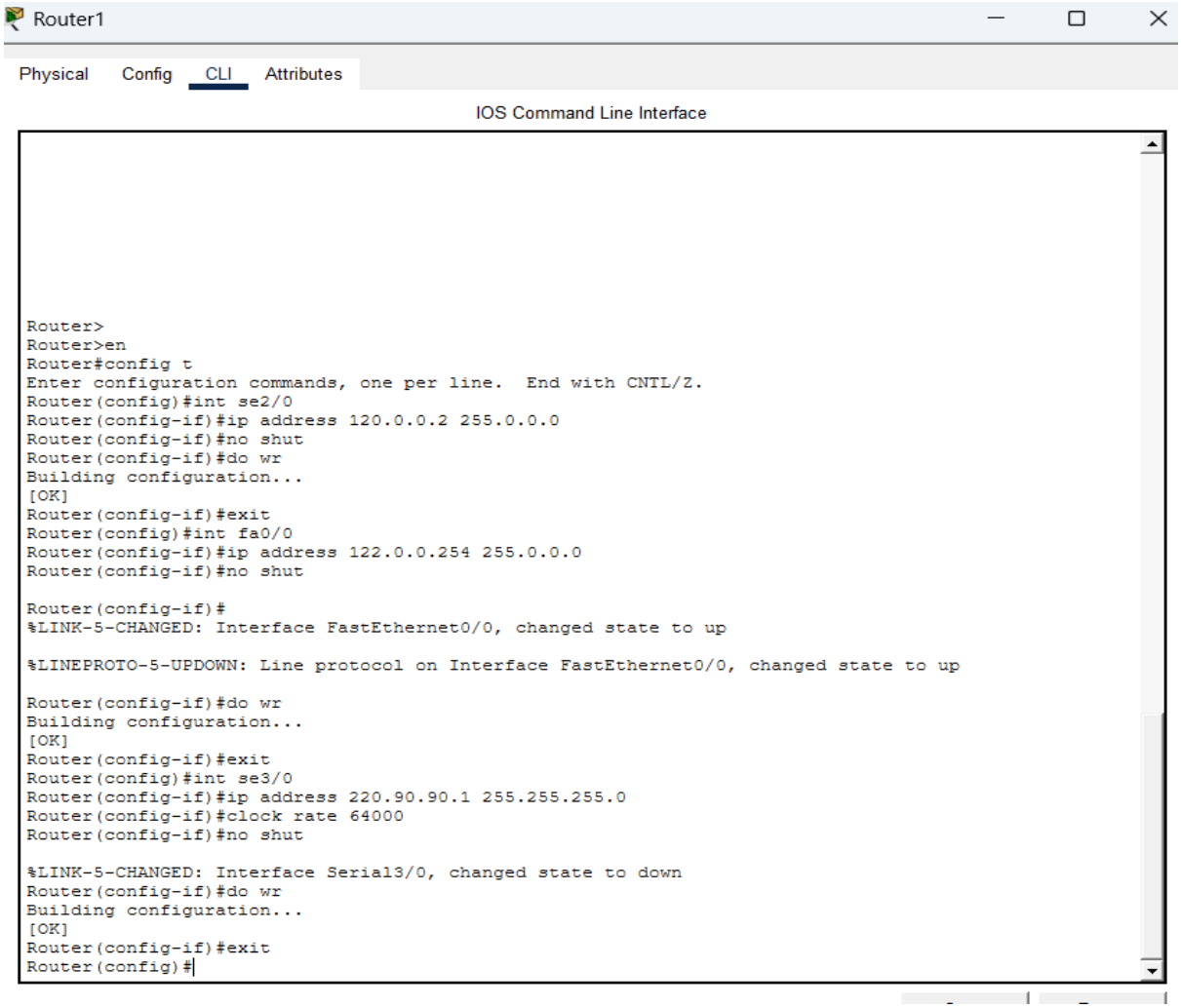


Figure: Router1 configuration

Router1 (RIP Configuration)

Router(config)#router rip
Router(config-router)#network 120.0.0.0
Router(config-router)#network 122.0.0.0
Router(config-router)#network 220.90.90.0
Router(config-router)#exit

```
Router(config)#router rip
Router(config-router)#network 120.0.0.0
Router(config-router)#network 122.0.0.0
Router(config-router)#network 220.90.90.0
Router(config-router)#exit
```

Router1 (OSPF Configuration)

Router(config)#router ospf 2
Router(config-router)#network 120.0.0.0 0.255.255.255 area 1
Router(config-router)#network 122.0.0.0 0.255.255.255 area 1

Router(config-router)#network 220.90.90.0 0.0.0.255 area 1
Router(config-router)#exit

```
Router(config)#router ospf 2
Router(config-router)#network 120.0.0.0 0.255.255.255 area 1
Router(config-router)#network 122.0.0.0 0.255.255.255 area 1
Router(config-router)#network 220.90.90.0 0.0.0.255 area 1
Router(config-router)#exit
```

Router1 (EIGRP Configuration)

Router(config)#router eigrp 1
Router(config-router)#network 120.0.0.0 0.255.255.255
Router(config-router)#network 122.0.0.0 0.255.255.255
Router(config-router)#network 220.90.90.0 0.0.0.255
Router(config-router)#exit

```
Router(config)#router eigrp 1
Router(config-router)#network 120.0.0.0 0.255.255.255
Router(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 1: Neighbor 120.0.0.1 (Serial2/0) is up: new adjacency

Router(config-router)#network 122.0.0.0 0.255.255.255
Router(config-router)#network 220.90.90.0 0.0.0.255
Router(config-router)#exit
```

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
% Unknown command or computer name, or unable to find computer address

Router>en
Router#config
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#network 120.0.0.0
Router(config-router)#network 122.0.0.0
Router(config-router)#network 220.90.90.0
Router(config-router)#exit
Router(config)#router ospf 1
Router(config-router)#network 120.0.0.0 0.255.255.255 area 1
Router(config-router)#net
01:24:59: %OSPF-5-ADJCHG: Process 1, Nbr 190.110.0.254 on Serial2/0 from LOADING to FULL, Loading
Done

% Incomplete command.
Router(config-router)#exit
Router(config)#router ospf 2
Router(config-router)#network 120.0.0.0 0.255.255.255 area 1
Router(config-router)#network 122.0.0.0 0.255.255.255 area 1
Router(config-router)#network 220.90.90.0 0.0.0.255 area 1
Router(config-router)#exit
Router(config)#router eigrp 1
Router(config-router)#netowrk 120.0.0.0 0.255.255.255
                          ^
% Invalid input detected at '^' marker.

Router(config-router)#exit
Router(config)#router eigrp 1
Router(config-router)#network 120.0.0.0 0.255.255.255
Router(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 1: Neighbor 120.0.0.1 (Serial2/0) is up: new adjacency

Router(config-router)#network 122.0.0.0 0.255.255.255
Router(config-router)#network 220.90.90.0 0.0.0.255
Router(config-router)#exit
Router(config)#
%LINK-5-CHANGED: Interface Serial3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/0, changed state to up

Router(config)#exit
Router#
```

Figure: Router1 configuration with dynamic routing protocols OSPF, RIP and EIGRP

For Router 2:

se2/0

Router>en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int se 2/0
Router(config-if)#ip address 220.90.90.2 255.255.255.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit

fa0/0

Router(config)#int fa0/0
Router(config-if)#ip address 222.0.0.254 255.255.255.0
Router(config-if)#no shut
Router(config-if)#do wr
Router(config-if)#exit

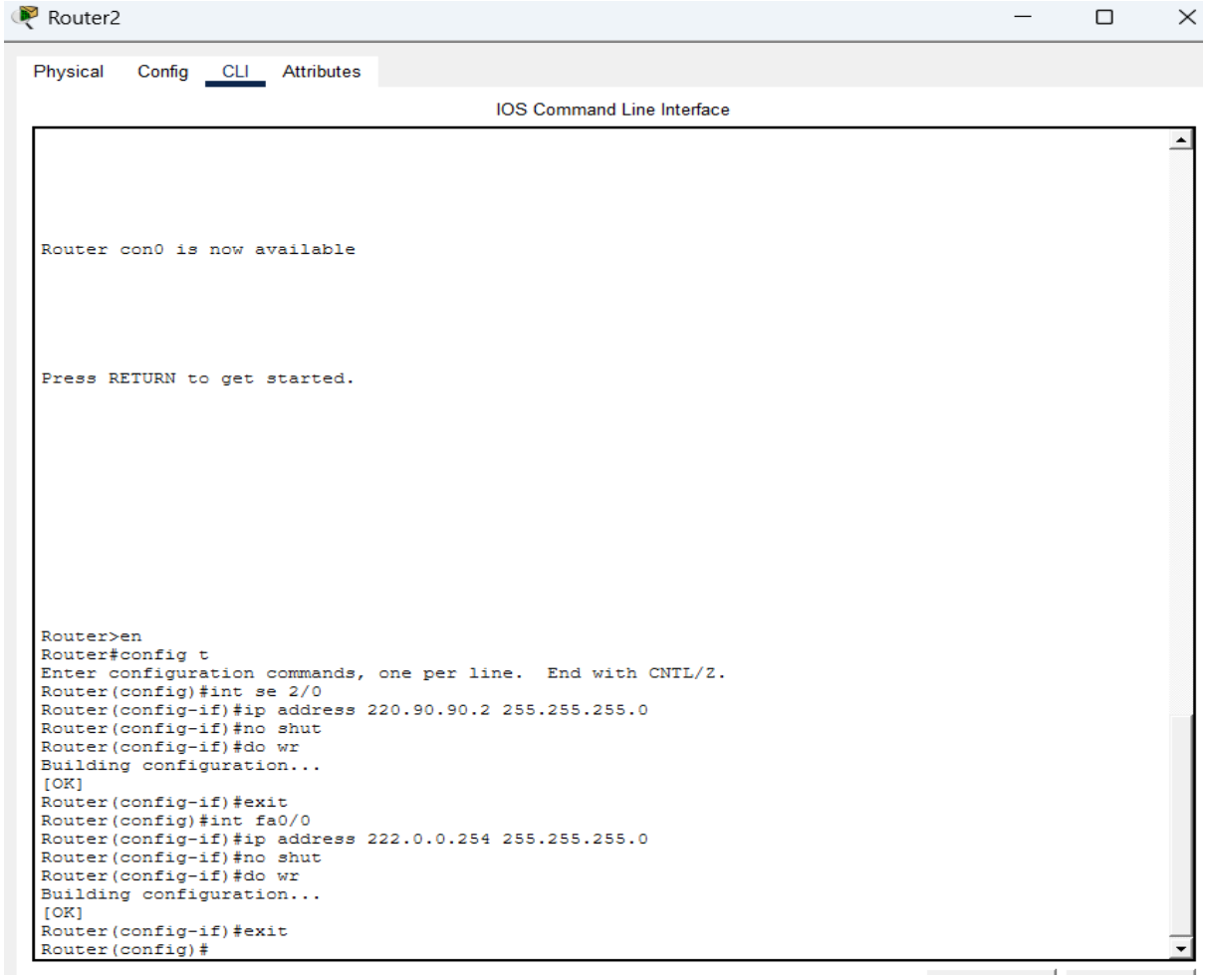


Figure: Router2 configuration

Router2 (RIP Configuration)

Router(config)#router rip
Router(config-router)#network 220.90.90.0
Router(config-router)#network 222.0.0.0
Router(config-router)#exit

```
Router(config)#router rip
Router(config-router)#network 220.90.90.0
Router(config-router)#network 222.0.0.0
Router(config-router)#exit
```

Router2 (OSPF Configuration)

Router(config)#router ospf 3
Router(config-router)#network 220.90.90.0 0.0.0.255 area 1
Router(config-router)#network 222.0.0.0 0.0.0.255 area 1
Router(config-router)#exit

```
Router(config)#router ospf 3
Router(config-router)#network 220.90.90.0 0.0.0.255 area 1
Router(config-router)#network 222.0.0.0 0.0.0.255 area 1
Router(config-router)#exit
```

Router2 (EIGRP Configuration)

Router(config)#router eigrp 1
Router(config-router)#network 220.90.90.0 0.0.0.255
Router(config-router)#network 222.0.0.0 0.0.0.255
Router(config-router)#exit

```
Router(config)#router eigrp 1
Router(config-router)#network 220.90.90.0 0.0.0.255
Router(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 1: Neighbor 220.90.90.1 (Serial2/0) is up: new adjacency

Router(config-router)#network 222.0.0.0 0.0.0.255
Router(config-router)#exit
```

Router2

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int se 2/0
Router(config-if)#ip address 220.90.90.2 255.255.255.0
Router(config-if)#no shut
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#int fa0/0
Router(config-if)#ip address 222.0.0.254 255.255.255.0
Router(config-if)#no shut
Router(config-if)#do wr
Building configuration...
[OK]
Router(config-if)#exit
Router(config)#router rip
Router(config-router)#network 220.90.90.0
Router(config-router)#network 222.0.0.0
Router(config-router)#exit
Router(config)#router ospf 3
Router(config-router)#network 220.90.90.0 0.0.0.255 area 1
Router(config-router)#network
01:49:37: %OSPF-5-ADJCHG: Process 3, Nbr 122.0.0.254 on Serial2/0 from LOADING to FULL, Loading
Done

% Incomplete command.
Router(config-router)#exit
Router(config)#router ospf 3
Router(config-router)#network 220.90.90.0 0.0.0.255 area 1
Router(config-router)#network 222.0.0.0 0.0.0.255 area 1
Router(config-router)#exit
Router(config)#router eigrp 1
Router(config-router)#network 220.90.90.0 0.0.0.255
Router(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 1: Neighbor 220.90.90.1 (Serial2/0) is up: new adjacency

Router(config-router)#network 222.0.0.0 0.0.0.255
Router(config-router)#exit
Router(config)#
```

Figure: Router2 configuration with dynamic routing protocols OSPF, RIP and EIGRP

Device Configuration

For PC 0

PC0

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

DHCP Static

IPv4 Address 190.110.0.1

Subnet Mask 255.255.0.0

Default Gateway 190.110.0.254

DNS Server

Figure: PC Configuration of PC0

For PC 1

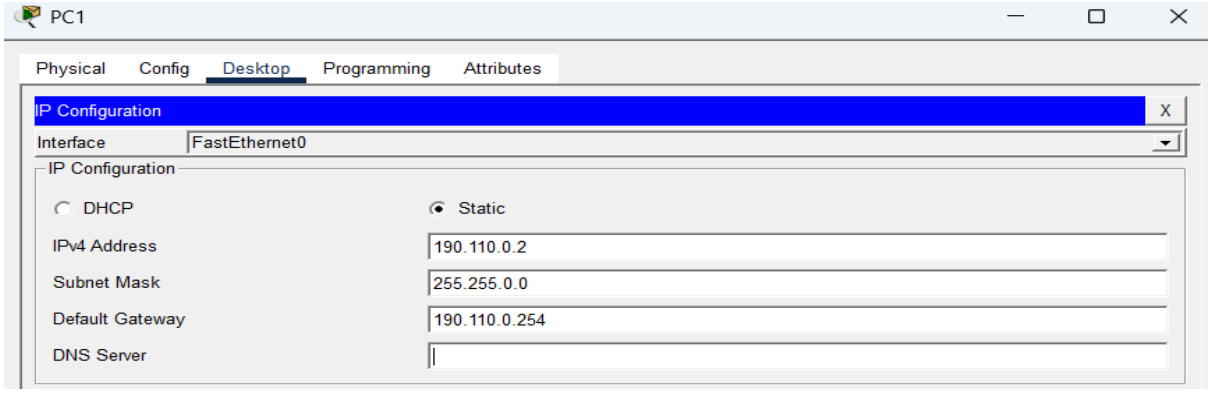


Figure: PC Configuration of PC1

For PC 2

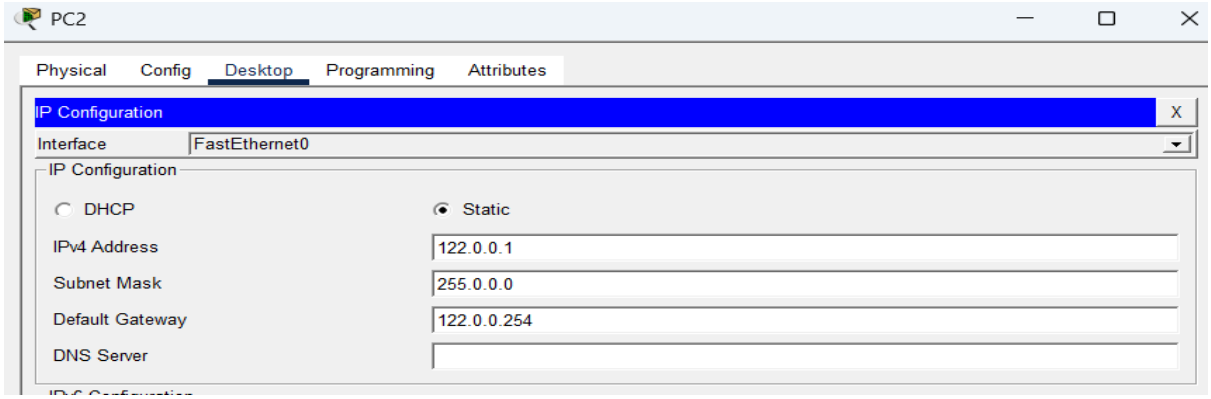


Figure: PC Configuration of PC2

For PC 3

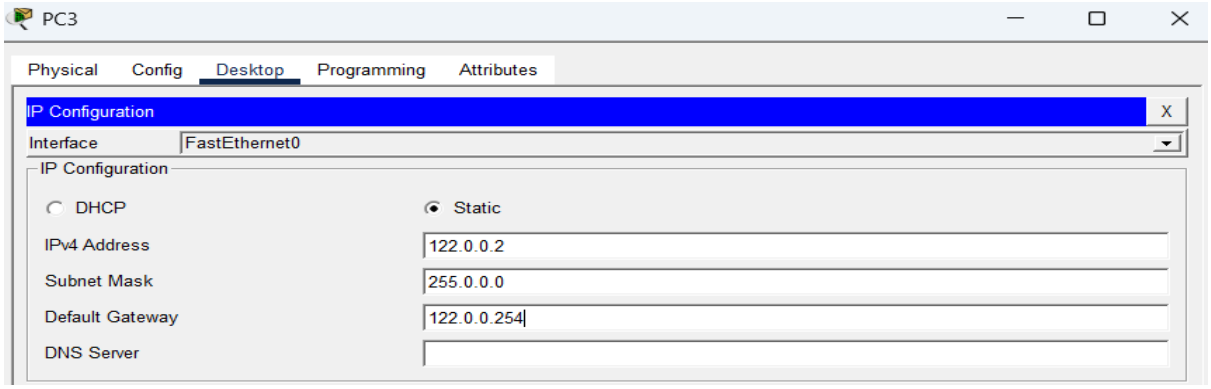


Figure: PC Configuration of PC3

For PC 4

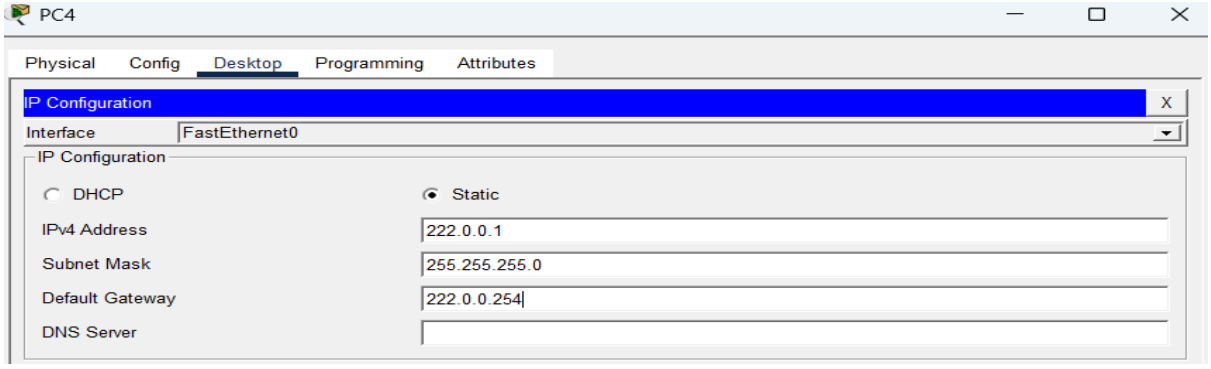


Figure: PC Configuration of PC4

For PC 5

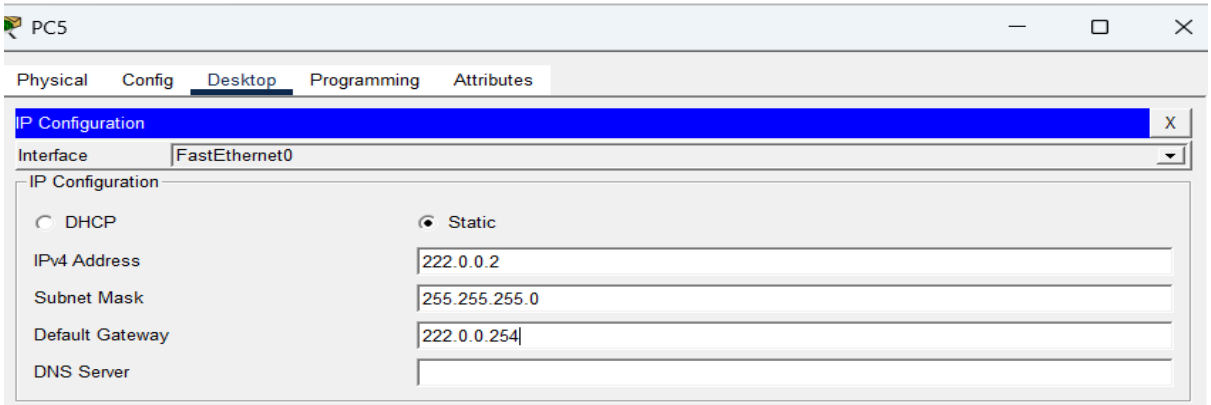


Figure: PC Configuration of PC5

Ping Communication:

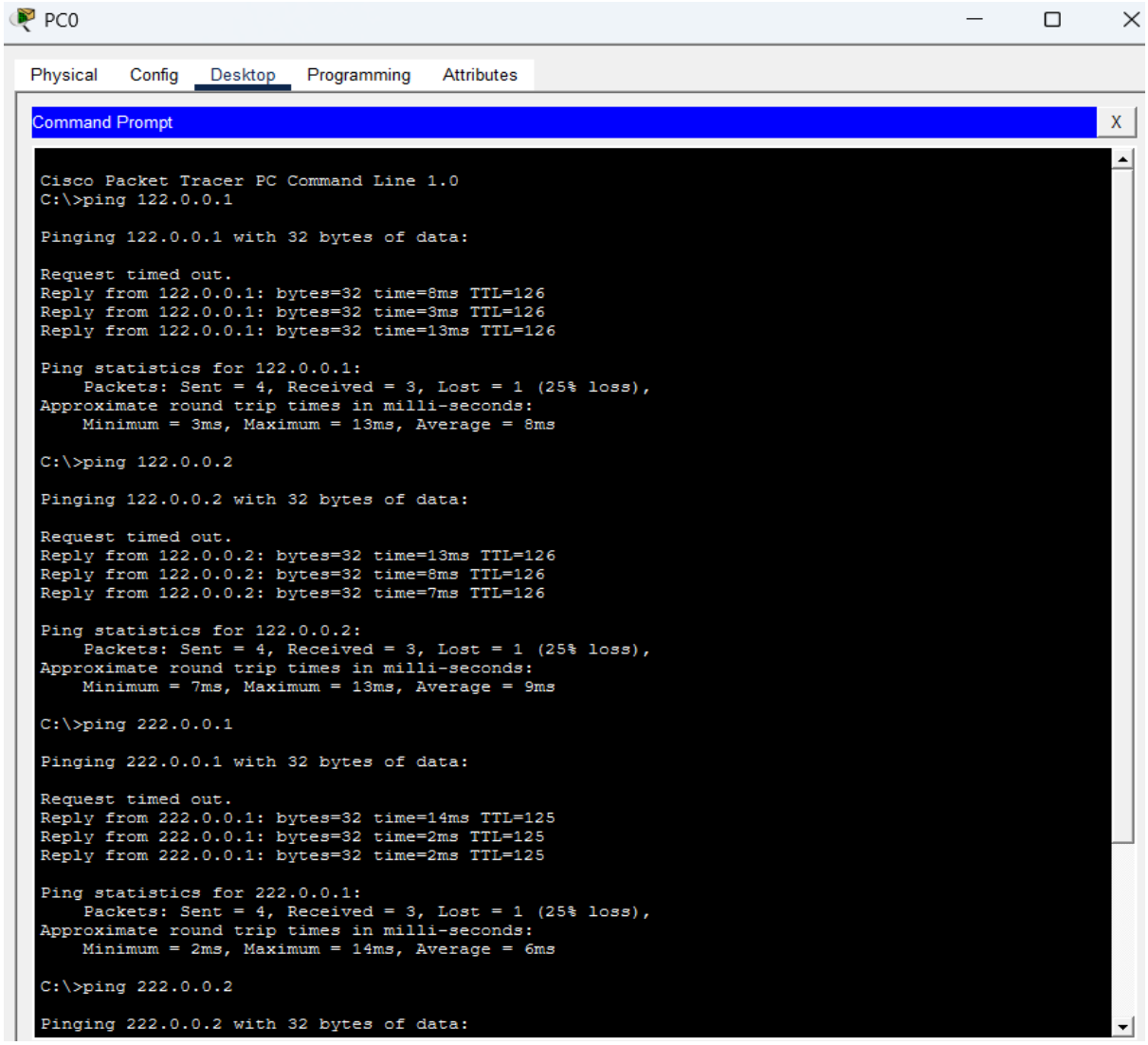


Figure: Ping Command

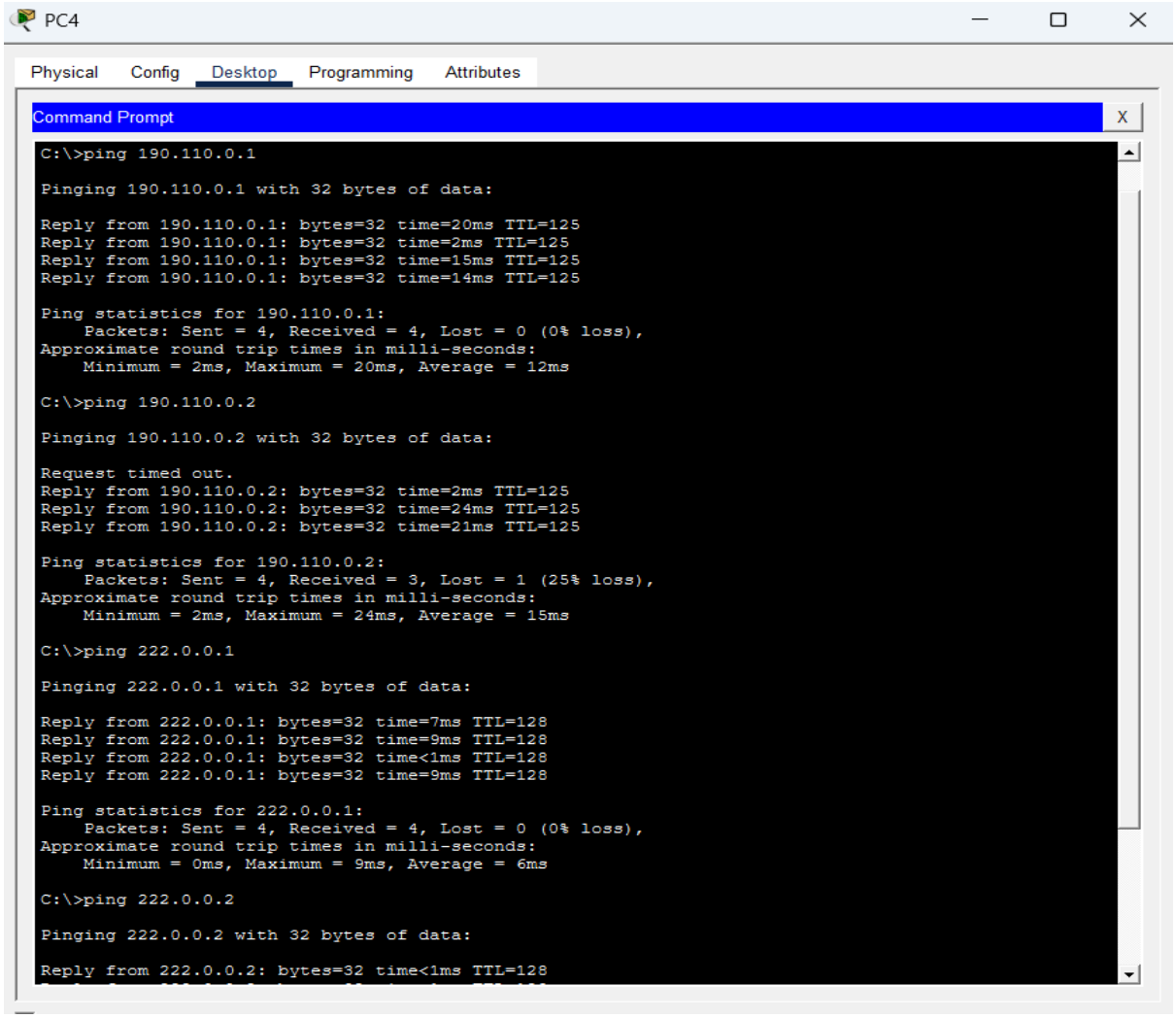
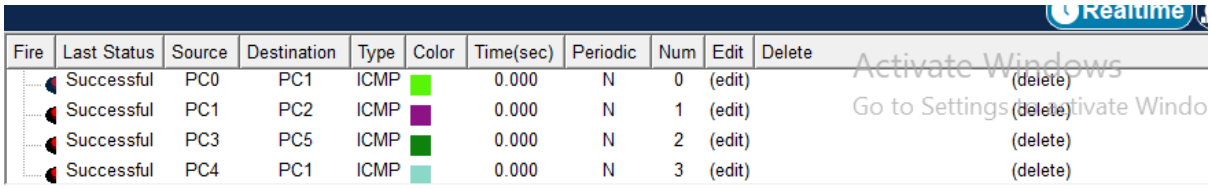


Figure : Ping Command

Test Results Between PCs



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC1	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC1	PC2	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC3	PC5	ICMP		0.000	N	2	(edit)	(delete)
	Successful	PC4	PC1	ICMP		0.000	N	3	(edit)	(delete)

Figure: Ping Test Results Between PCs